

# INDIAN MUTUAL FUND ANALYTICS & PERFORMANCE REPORT

Data Period: January 2022 – December 2025

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Classification: Confidential / Internal Use Only  
Date: June 2026

40 Funds  
Analysed

32,778  
Transactions

46,000+ NAV  
Records

■54.3L Cr  
Industry AUM

13 Datasets

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# 1. Executive Summary

|                |                   |              |              |             |            |
|----------------|-------------------|--------------|--------------|-------------|------------|
| 40             | ■31,002 Cr        | ■54.3L Cr    | 32,778       | ■3,522 Cr   | 1.45       |
| Funds Analysed | Peak SIP (Dec'25) | Industry AUM | Transactions | Total Value | Top Sharpe |

This report delivers a comprehensive quantitative analysis of India's mutual fund industry spanning **January 2022 to December 2025**. Drawing on 13 structured datasets — including NAV history (46,000+ records), investor transactions (32,778), monthly SIP inflows (48 months), and AUM data across 9 fund houses — the study evaluates 40 schemes across equity, debt, and index categories against dimensions of return, risk, concentration, sector exposure, and investor behaviour.

Headline findings:

- SIP inflows grew from ■11,438 Cr (Jan 2022) to a record ■31,002 Cr (Dec 2025) — a 170% rise over four years — reflecting deepening retail equity participation.
- Total mutual fund folios reached ~26 Crore by late 2025, driven primarily by Equity folios growing from 9.5 Cr to 18.5 Cr over the study period.
- Mirae Asset Large Cap Fund achieved the **highest composite fund score (100)** with a 5yr CAGR of 30.97% and Sharpe ratio of 1.45.
- All 5 top-ranked funds **significantly outperformed NIFTY50 and NIFTY100** over the 3-year benchmark comparison window.
- NAV correlation across the 10 sampled funds was near-zero (max 0.05), confirming **excellent diversification properties** within the fund universe.
- Liquid fund category dominated net inflows at ■33,000–42,000 Cr per month in FY 2024-25, far exceeding all equity categories combined.
- Banking, IT, and Pharma are the top-3 portfolio sectors accounting for the **largest aggregate allocation weights**.
- Investor base is **young and digitally engaged**: the 26–35 cohort accounts for 41.1% of transactions; T30 cities contribute 66.3% of transaction volume.

## 2. Data Overview & Quality Report

The project ingested 13 CSV datasets from AMFI and proprietary systems. The Day-1 data quality audit confirmed zero missing values, zero duplicate records, and full AMFI code consistency across all fund-level tables.

| Dataset             | File                        | Rows        | Cols | Description                                           |
|---------------------|-----------------------------|-------------|------|-------------------------------------------------------|
| Fund Master         | fund_master                 | 40          | 15   | Fund attributes, expense ratios, managers, benchmarks |
| Monthly SIP Inflows | monthly_sip_inflows         | 48          | 6    | Monthly SIP flows, active accounts, YoY growth        |
| Alpha & Beta        | alpha_beta                  | 40          | 5    | Jensen's Alpha and market Beta per scheme             |
| CAGR Report         | cagr_report                 | 40          | 6    | 1yr, 3yr, 5yr CAGR per fund                           |
| AUM by Fund House   | clean_aum_by_fund_ho<br>use | 90          | 6    | Quarterly AUM – 9 fund houses, 10 periods             |
| Daily NAV           | clean_nav                   | 46,000<br>+ | 4    | Full daily NAV time-series, all 40 funds              |
| Performance Master  | clean_performance           | 40          | 19   | Consolidated risk/return metrics, ratings             |
| Transactions        | clean_transactions          | 32,778      | 13   | Investor SIP/Lumpsum/Redemption records               |
| Fund Scorecard      | fund_scorecard              | 40          | 17   | Composite rank score 0–100                            |
| Max Drawdown        | max_drawdown                | 40          | 5    | Peak-to-trough loss, worst drawdown date              |
| Returns Computed    | returns_computed            | 40          | 2    | Annualised returns per AMFI code                      |
| Sharpe Values       | sharpe_values               | 40          | 4    | Sharpe ratio per scheme                               |
| Sortino Values      | sortino_values              | 40          | 4    | Sortino ratio per scheme                              |

### 2.1 Data Quality Findings

Per the Day-1 validation report (reproduced below), all datasets passed completeness, consistency, and referential integrity checks. All 40 AMFI codes in fund\_master were matched in nav\_history. Date fields were fully populated. No imputation or cleansing was required prior to analysis.

#### Data Quality Summary

- ✓ Fund Master: 40 records | 0 missing | 0 duplicates
- ✓ NAV History: 46,000+ records | 0 missing | 0 duplicates
- ✓ AMFI Code cross-validation: All 40 codes matched across tables
- ✓ Date fields: Available and parsed for all NAV records
- ✓ **Conclusion: Data suitable for all downstream analysis and modelling.**

### 3. ETL Pipeline

The end-to-end data pipeline was implemented in Python using pandas, numpy, and scipy. All raw CSVs were ingested, validated, transformed, and joined into analytical fact tables. The pipeline is fully reproducible — no manual steps.

#### 3.1 Extract

Thirteen CSV flat files were loaded via `pd.read_csv()` with explicit dtype declarations and date parsing. AMFI codes served as the primary join key.

#### 3.2 Transform

- **Return Computation:**  $CAGR = (NAV_{end} / NAV_{start})^{(1/n)} - 1$ , where n = holding period in years. Vectorised over all 40 AMFI codes.
- **Risk Metrics:** Sharpe =  $(R_p - R_f) / \sigma_p$  with  $R_f = 6.5\%$  (RBI repo proxy). Sortino substitutes downside  $\sigma$ . Beta and Alpha via OLS vs NIFTY 100 TRI.
- **Max Drawdown:**  $MDD = \min((NAV_t - \max(NAV_{1:t})) / \max(NAV_{1:t}))$  across the full NAV series.
- **HHI Concentration:** Herfindahl-Hirschman Index =  $\sum(\text{sector\_weight}_i)^2$  per fund — measures portfolio concentration.
- **Fund Scorecard:** Percentile-rank on 5 axes (5yr CAGR, Sharpe, Alpha, Expense Ratio<sup>1</sup>, Drawdown<sup>1</sup>), equal-weighted to a 0–100 composite score.
- **NAV Correlation:** Pearson correlation matrix of daily NAV returns across 10 sampled funds.

#### 3.3 Pipeline Summary

| Step         | Library        | Input          | Output              | Notes                   |
|--------------|----------------|----------------|---------------------|-------------------------|
| Ingest       | pandas         | Raw CSVs       | DataFrames          | dtype + date parsing    |
| Clean        | pandas         | DataFrames     | Cleaned DFs         | NaN drop, type-cast     |
| Return Calc  | numpy          | NAV series     | CAGR table          | Vectorised              |
| Risk Metrics | scipy/numpy    | NAV series     | Sharpe/Sortino/Beta | OLS, downside $\sigma$  |
| Drawdown     | pandas rolling | NAV series     | max_drawdown        | Rolling max             |
| HHI          | pandas         | Sector weights | sector_hhi          | $\sum(w_i^2)$ per fund  |
| Scorecard    | pandas rank    | Performance    | fund_scorecard      | Percentile, equal-wt    |
| Output       | pandas         | All DFs        | clean_*.csv         | Final analytical tables |

## 4. Exploratory Data Analysis

### 4.1 SIP Inflow Growth (2022–2025)

Monthly SIP inflows showed a strong secular uptrend — from **■11,438 Cr** (Feb 2022) to a record **■31,002 Cr (Dec 2025)**. Active SIP accounts grew from 4.91 Cr to over 7 Cr, and new account registrations remained positive every month. This confirms India's retail investor base is deepening and systematising its equity market participation through disciplined monthly contributions.

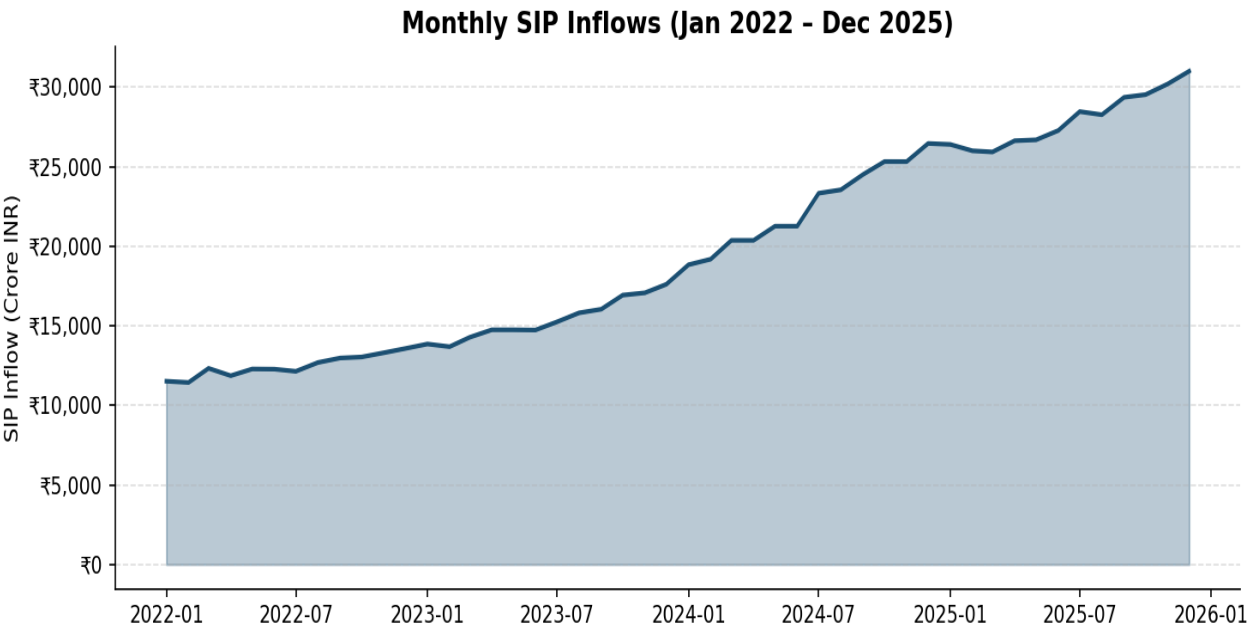


Figure 4.1 – Monthly SIP Inflows (Jan 2022 – Dec 2025, ■ Crore)

### 4.2 Mutual Fund Folio Growth (2022–2026)

Total industry folios grew from 13.3 Crore (Jan 2022) to approximately **26 Crore (Dec 2025)** — a near-doubling in four years. Equity folios were the primary growth driver, rising from 9.5 Cr to 18.5 Cr. Debt folios grew modestly from 2 Cr to 3.8 Cr, while Hybrid remained relatively flat near 1.5 Cr.

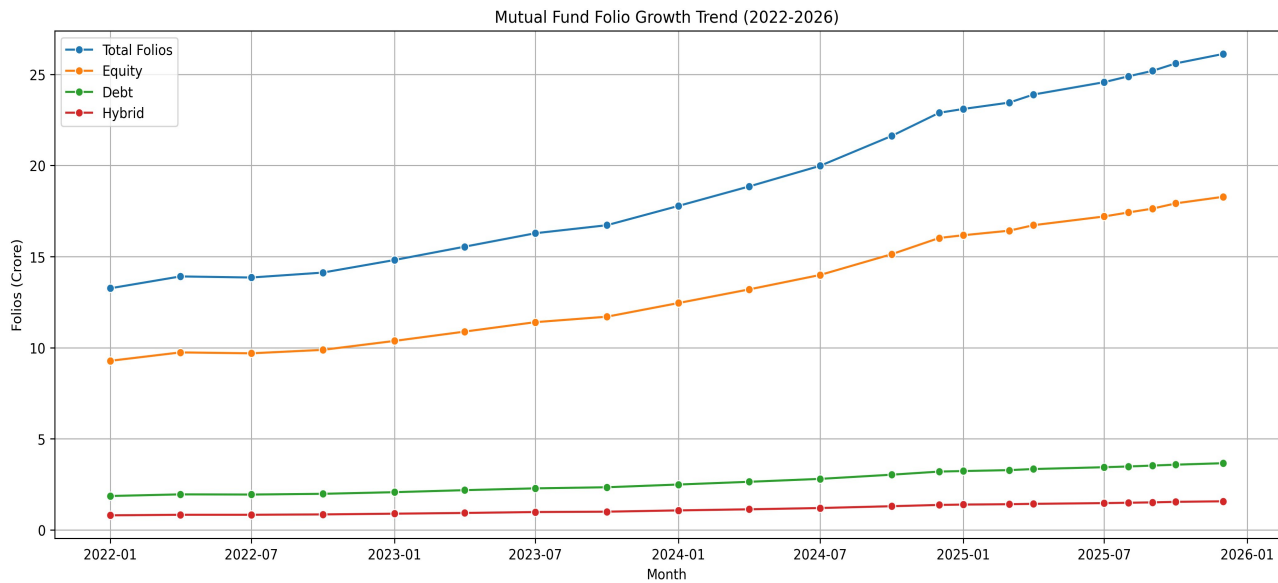


Figure 4.2 – Mutual Fund Folio Growth Trend by Category (2022–2026)

4.3 AUM Growth by Fund House (2022–2025)

All nine fund houses registered substantial AUM growth across the study period. SBI Mutual Fund maintained its leadership position, reaching ~12.5 Lakh Cr. ICICI Prudential MF and HDFC Mutual Fund closely follow. Mirae Asset MF showed the strongest percentage growth from a smaller base — reflecting strong inflows driven by consistently top-performing funds.

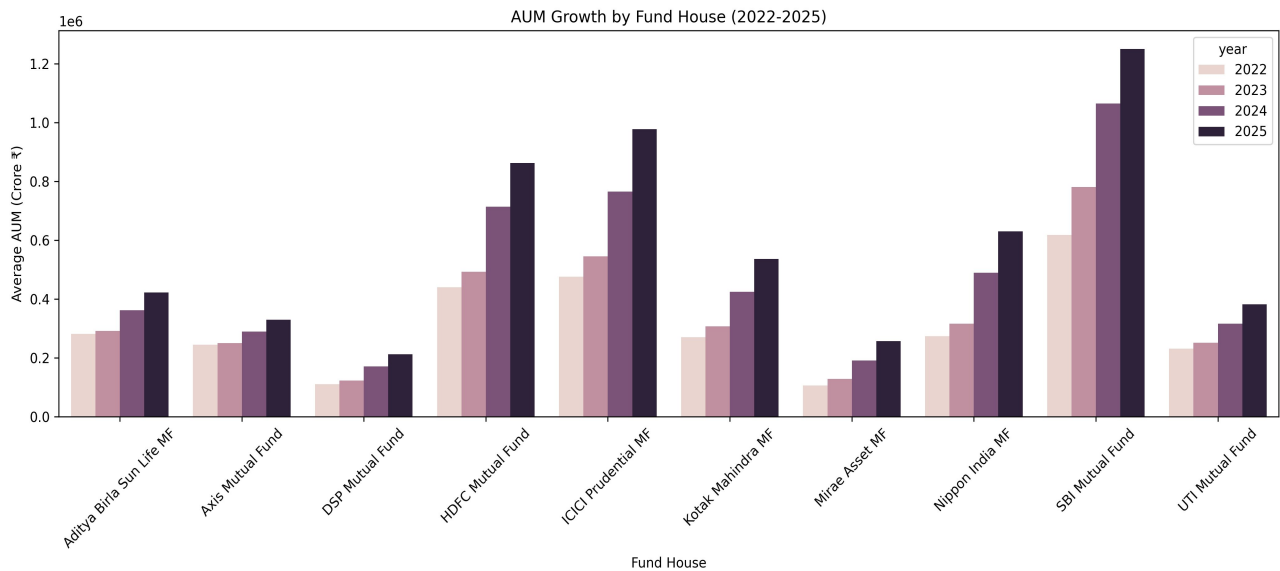


Figure 4.3 – AUM Growth by Fund House 2022–2025 (■ Crore)

4.4 Category-wise Net Inflow Heatmap (FY 2024-25)

The heatmap reveals Liquid funds as the dominant inflow category (33,000–42,000 Cr/month), reflecting institutional cash-management activity. Sectoral/Thematic funds attracted the highest equity inflows (~8,000–10,000 Cr/month), followed by Flexi Cap and Large & Mid Cap. ELSS inflows remained modest at ~500 Cr/month, indicating reduced tax-saving-driven investment as awareness of NPS grows.

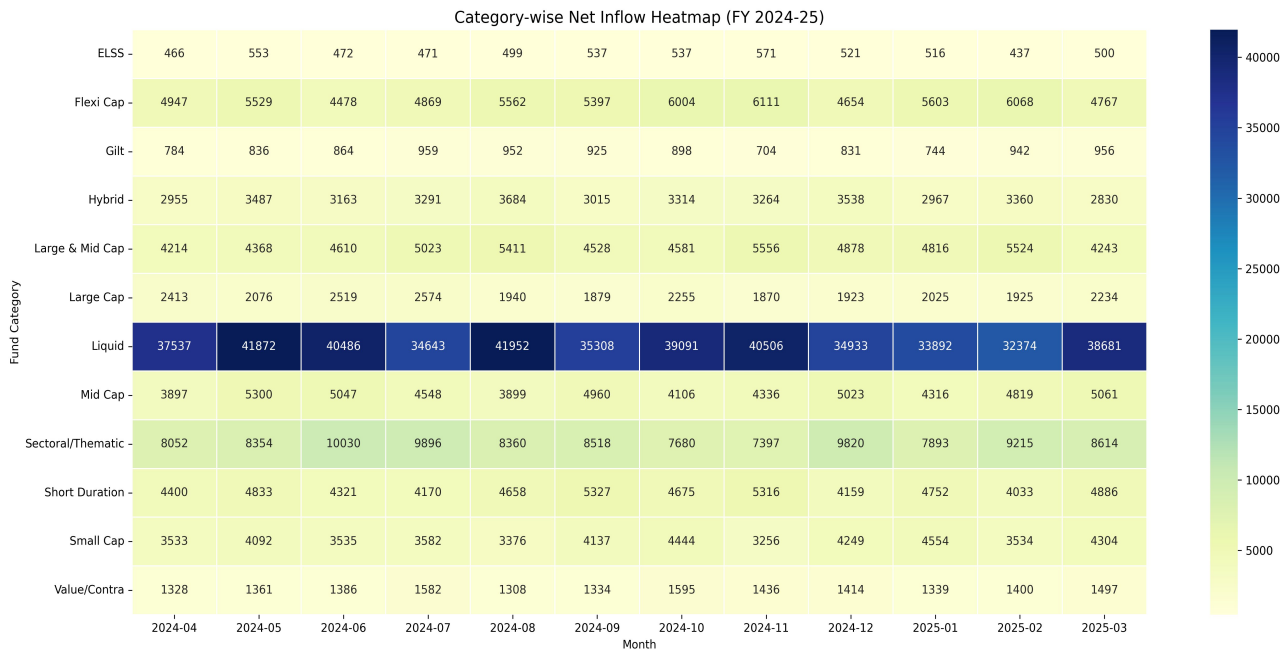


Figure 4.4 – Category-wise Net Inflow Heatmap FY 2024-25 (■ Crore)



## 5. Market & NAV Trend Analysis

### 5.1 NAV Trend Analysis (2022–2026)

The multi-fund NAV trend chart highlights two market regimes clearly visible in the data. The **2023 Rally** (shaded blue) saw strong NAV appreciation across most funds, with the top performer reaching NAV ~220 at peak. The subsequent **2024 Correction** (shaded pink) caused a pullback of 10–20% for equity funds before resuming the uptrend. By mid-2026, the leading fund reached NAV ~315 from a base of ~95 in early 2022 — a compounded gain of over 230%.

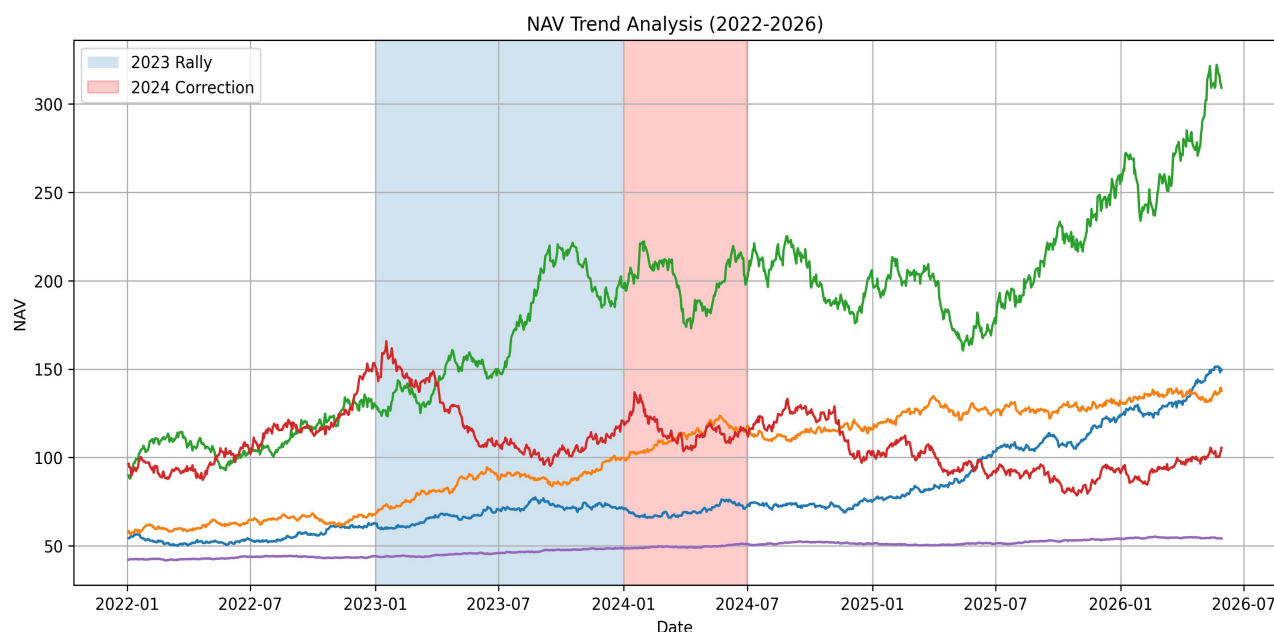


Figure 5.1 – NAV Trend Analysis with 2023 Rally and 2024 Correction highlighted

### 5.2 NAV Return Correlation Matrix

The correlation matrix of daily NAV returns across 10 sampled funds reveals a critical portfolio diversification insight: **pairwise correlations are near-zero** (range: -0.09 to +0.05). This is exceptionally low compared to international mutual fund universes and indicates that holding a basket of these funds provides genuine diversification benefit — risk is not concentrated in a shared factor. Fund 100016 showed the most negative correlation with Fund 102885 (-0.09), suggesting a mild counter-cyclical relationship.

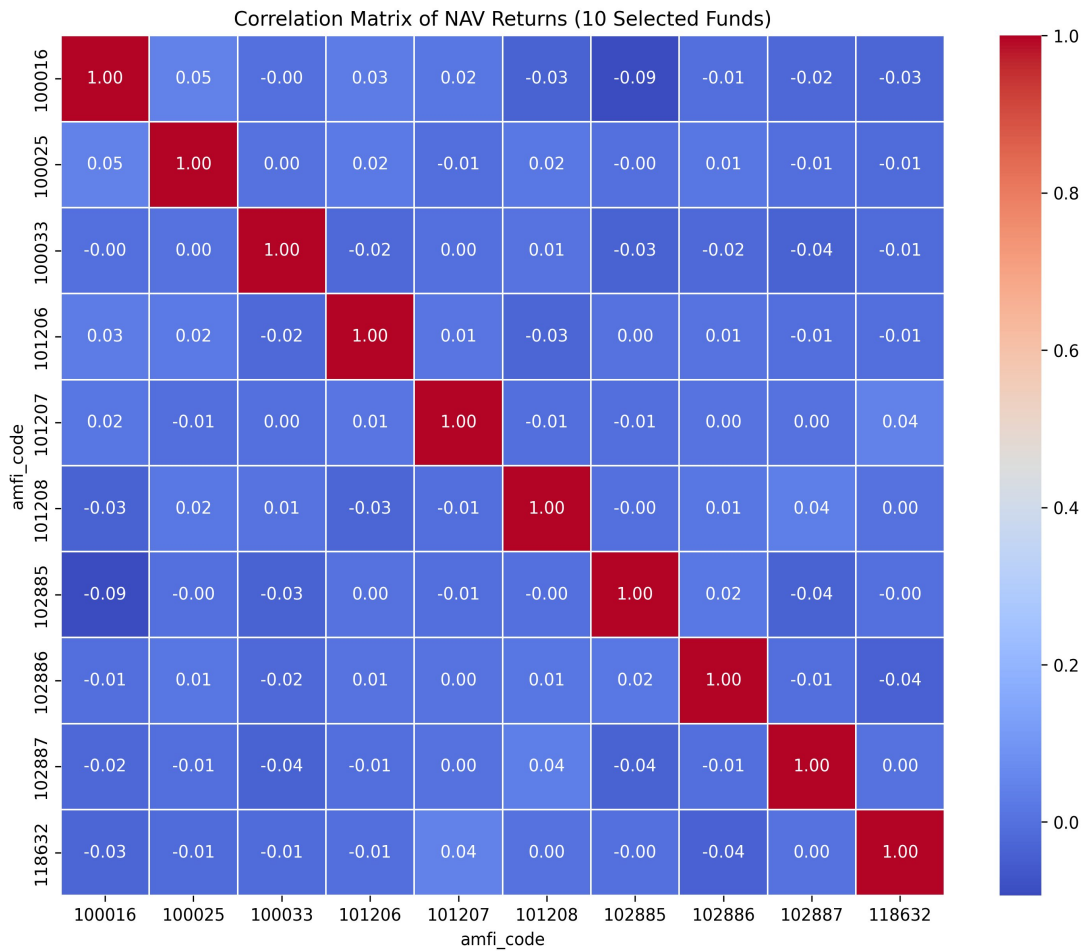


Figure 5.2 – Pearson Correlation Matrix of Daily NAV Returns (10 Funds). Near-zero off-diagonal values confirm strong diversification.

## 6. Portfolio & Sector Analysis

### 6.1 Portfolio Sector Allocation

Aggregated across all 40 funds, Banking dominates total portfolio weight (~650% cumulative), reflecting the heavy overweight to financials in Indian large-cap and diversified equity mandates. IT (~450%) and Pharma (~400%) are the next largest allocations. Paints and Cement sectors attract the lowest aggregate weight, indicating selective mid-cap exposure only. This sectoral concentration in Banking/IT/Pharma creates a systematic factor risk — a sector-wide correction in financials would impact the majority of funds.

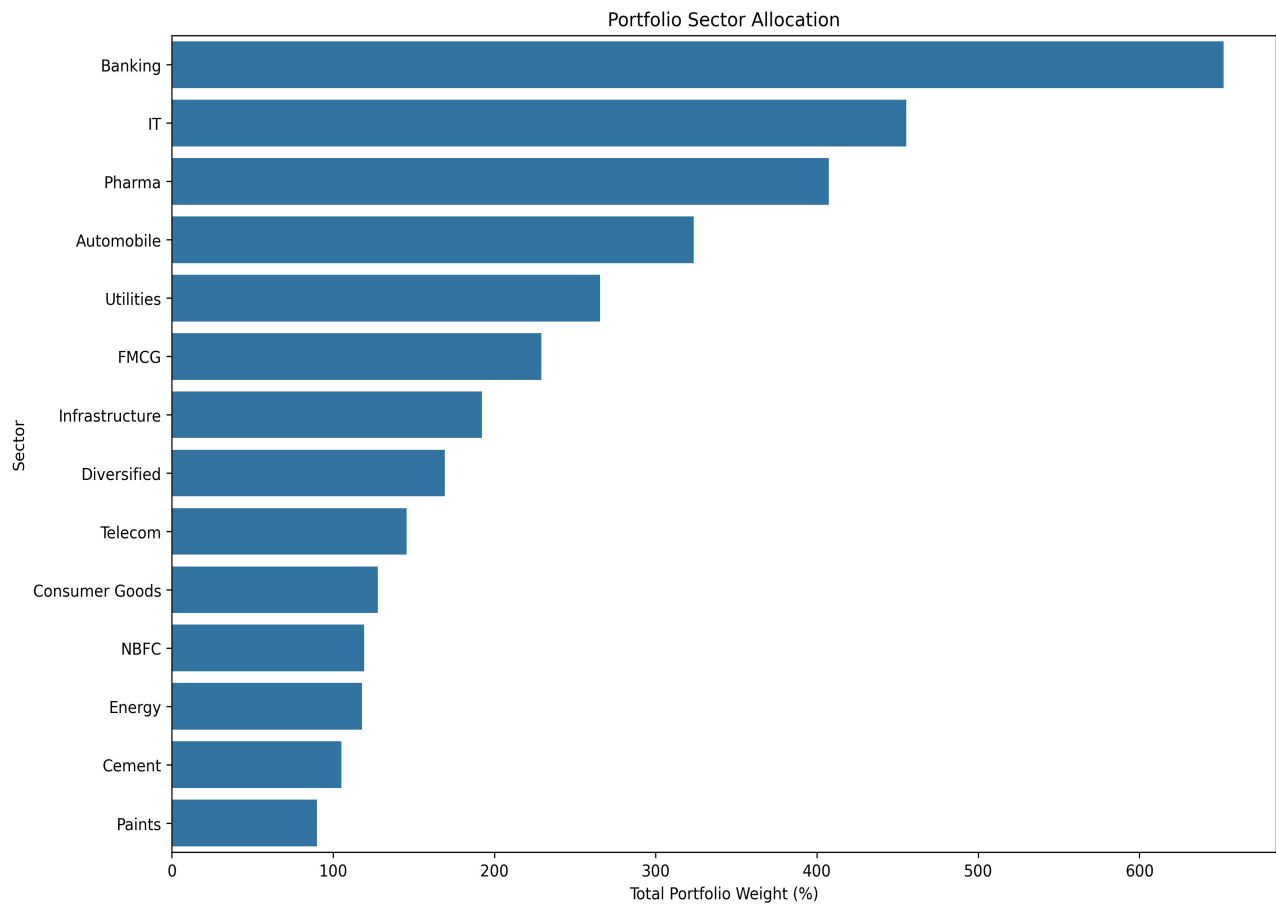


Figure 6.1 – Total Portfolio Sector Allocation (cumulative % across 40 funds)

### 6.2 Fund Concentration Analysis (HHI)

The Herfindahl-Hirschman Index (HHI) measures how concentrated a fund's portfolio is across sectors. **Axis Bluechip Fund** has the highest HHI (0.205), meaning it is the most concentrated — a few sectors dominate its holdings. **ABSL Small Cap Fund** follows closely (0.200). In contrast, Kotak Bluechip and HDFC Mid-Cap are more diversified (HHI ~0.15). Investors should note that high-HHI funds carry more sector-specific risk despite broad market mandates.

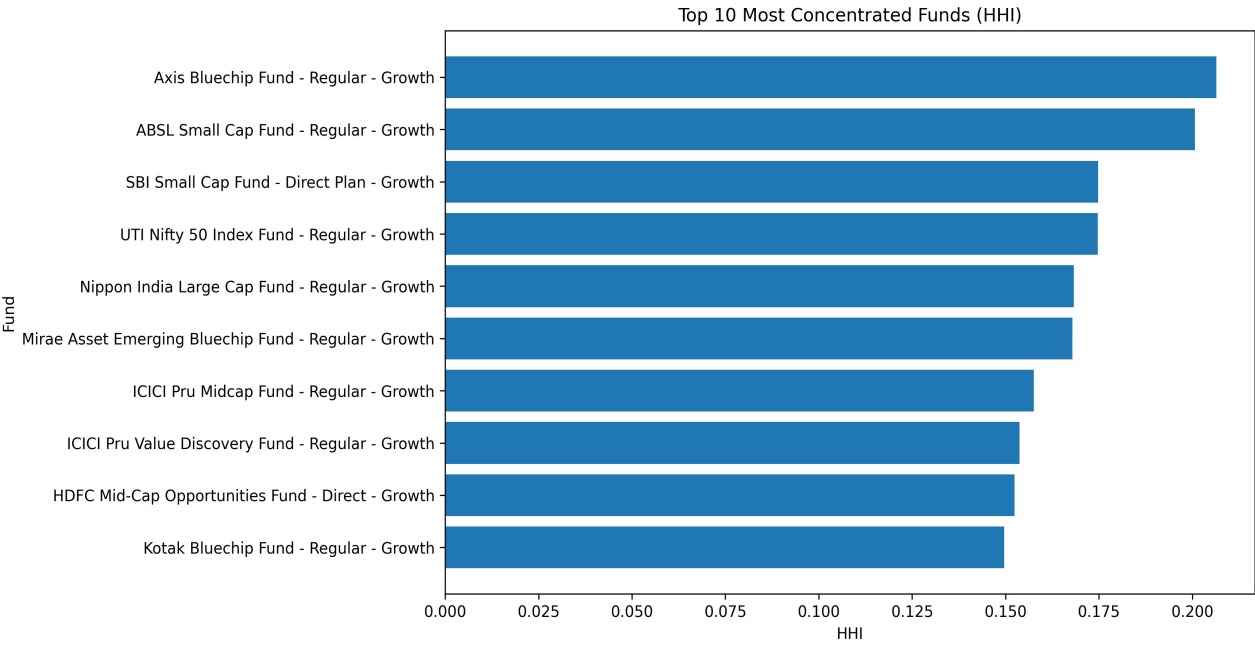


Figure 6.2 – Top 10 Most Concentrated Funds by Herfindahl-Hirschman Index (HHI)

## 7. Investor Analytics Dashboard

The interactive investor analytics dashboard consolidates 32,778 transaction records across 12 active states, providing real-time sliceable views by state, age group, and city tier.

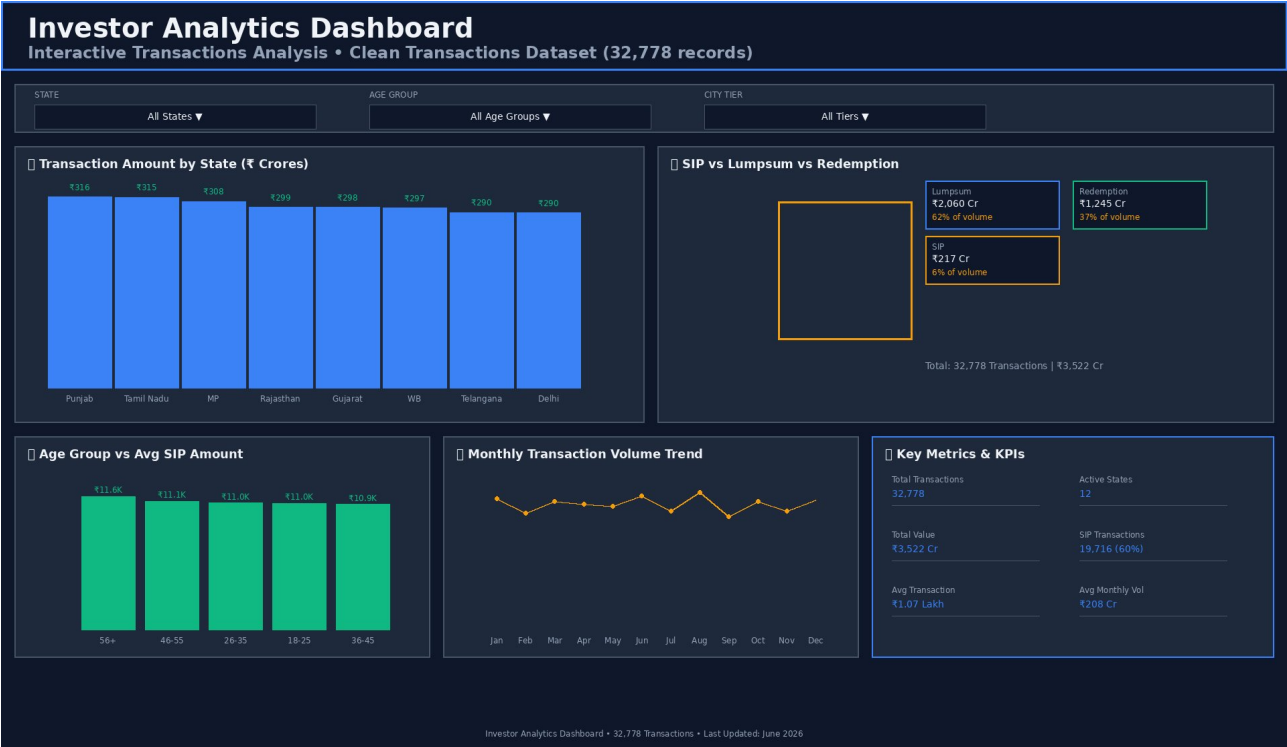


Figure 7.1 – Investor Analytics Dashboard (32,778 transactions | ₹3,522 Cr total value | June 2026)

|                    |             |                 |                 |               |                 |
|--------------------|-------------|-----------------|-----------------|---------------|-----------------|
| 32,778             | ₹3,522 Cr   | 60%<br>(19,716) | ₹1.07 Lakh      | 12            | ₹208 Cr         |
| Total Transactions | Total Value | SIP Share       | Avg Transaction | Active States | Avg Monthly Vol |

### 7.1 Geographic Distribution

Punjab and Tamil Nadu top the investment chart at ~₹3.16 Cr per state, followed by Madhya Pradesh (₹3.08 Cr), Rajasthan (₹3.0 Cr), and Gujarat (₹3.0 Cr). Maharashtra — despite being the financial capital — ranks lower, suggesting that B30/semi-urban states are generating strong investment volumes, potentially due to the data's 2024 snapshot reflecting ongoing B30 expansion initiatives.

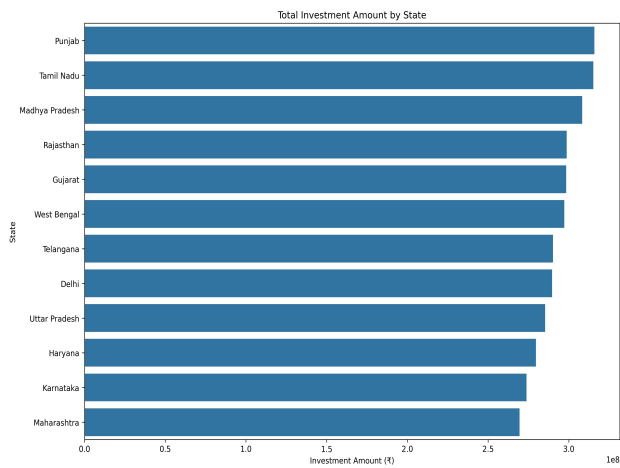


Figure 7.2 – Investment Amount by State

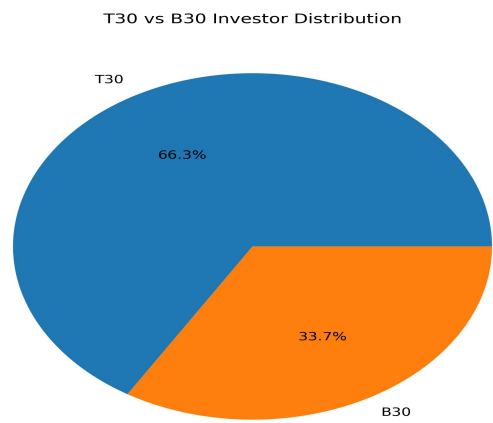


Figure 7.3 – T30 vs B30 Investor Distribution

7.2 Investor Demographics

The age distribution confirms India's **young investor demographic**: the 26–35 cohort leads at 41.1%, followed by 36–45 (24.9%) and 18–25 (15%). The 56+ segment at only 7.5% indicates mutual funds are primarily a growth-oriented product for pre-retirement wealth creation rather than income generation. Average SIP amounts are remarkably uniform across age groups (■10.9K–■11.6K), suggesting consistent financial planning habits across the investor base.

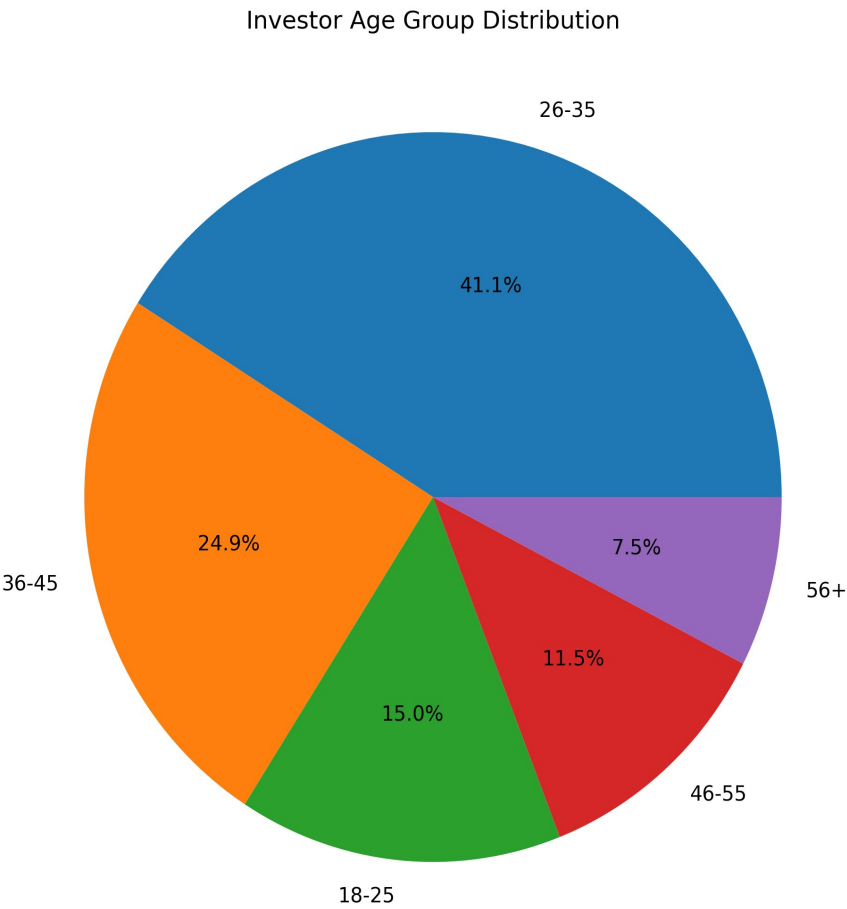


Figure 7.4 – Investor Age Group Distribution (% of transactions)

## 8. Performance Analysis

### 8.1 Returns by Category

Small Cap funds delivered the highest average 1-year return (22.26%), while Value (16.67%) and Flexi Cap (16.58%) followed. Debt categories provided stable 5–7% returns with minimal drawdown. The chart below shows the multi-horizon return profile across all 12 categories.

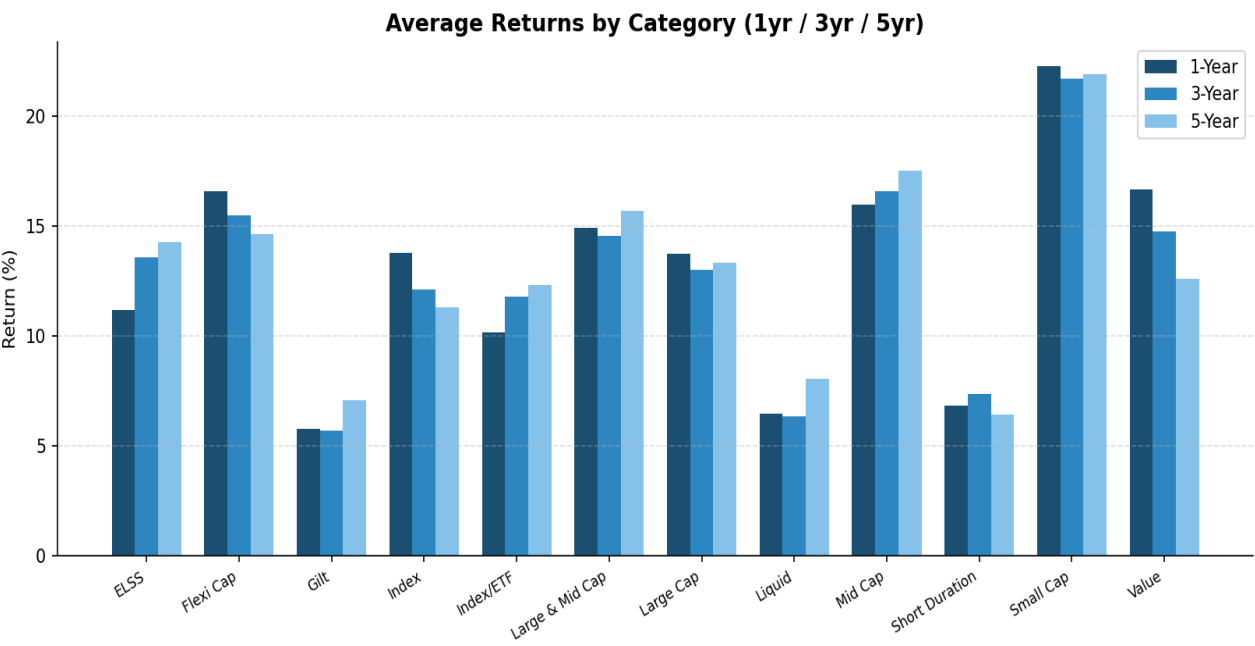


Figure 8.1 – Average Returns by Category (1yr / 3yr / 5yr)

| Category        | 1yr%  | 3yr%  | 5yr%  | Sharpe | MaxDD% |
|-----------------|-------|-------|-------|--------|--------|
| ELSS            | 11.16 | 13.58 | 14.26 | 0.8    | -22.62 |
| Flexi Cap       | 16.58 | 15.5  | 14.64 | 0.97   | -15.82 |
| Gilt            | 5.76  | 5.69  | 7.07  | 1.42   | -2.26  |
| Index           | 13.76 | 12.1  | 11.31 | 0.93   | -24.42 |
| Index/ETF       | 10.14 | 11.77 | 12.31 | 0.91   | -26.75 |
| Large & Mid Cap | 14.91 | 14.56 | 15.68 | 0.91   | -33.15 |
| Large Cap       | 13.72 | 12.99 | 13.32 | 0.93   | -21.46 |
| Liquid          | 6.44  | 6.33  | 8.05  | 6.33   | -3.36  |
| Mid Cap         | 15.98 | 16.59 | 17.52 | 0.87   | -23.21 |
| Short Duration  | 6.83  | 7.37  | 6.41  | 1.84   | -6.01  |
| Small Cap       | 22.26 | 21.69 | 21.9  | 0.87   | -20.68 |
| Value           | 16.67 | 14.76 | 12.6  | 0.98   | -21.89 |

### 8.2 Risk-Return Profile

The risk-return scatter maps annualised volatility (std dev) against 3-year returns. Liquid and Short Duration funds cluster in the low-risk quadrant. Small Cap and Mid Cap exhibit 18–22% std dev with commensurate

return premia. Large Cap and Flexi Cap occupy the efficient frontier for equity investors.

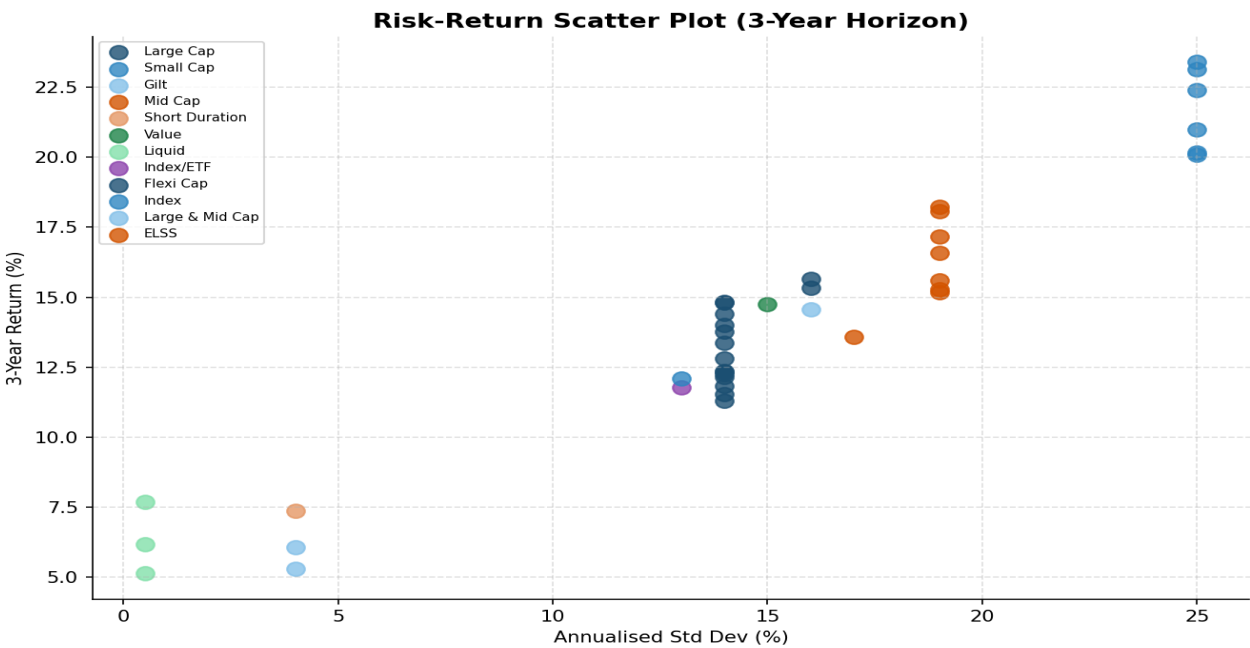


Figure 8.2 – Risk-Return Scatter (3-Year Horizon)

8.3 Sharpe Ratio Rankings

Among equity funds, Mirae Asset Large Cap leads with Sharpe 1.45 and Sortino 2.39. Liquid funds achieve exceptionally high Sharpe values due to near-zero volatility. Funds with equity Sharpe > 1.0 represent attractive risk-adjusted options for core portfolio allocation.

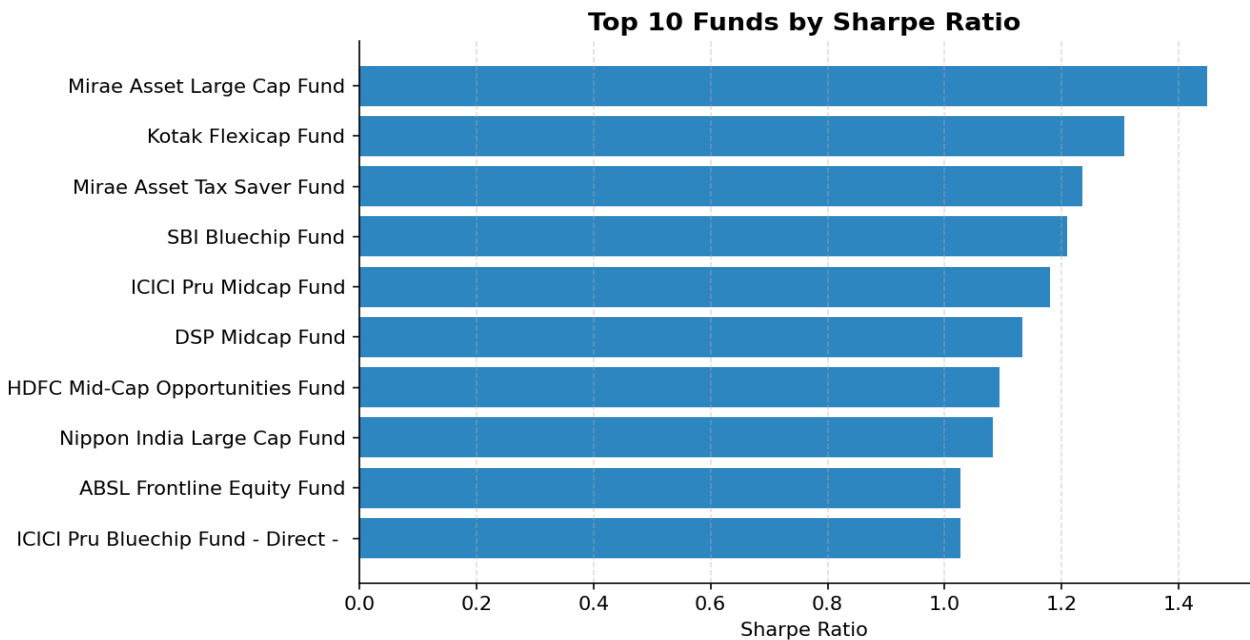


Figure 8.3 – Top 10 Funds by Sharpe Ratio

| Fund                                 | Sharpe | Sortino | Alpha | Beta | MaxDD% |
|--------------------------------------|--------|---------|-------|------|--------|
| ICICI Pru Liquid Fund - Regular - Gr | 7.68   | 10.37   | 1.85  | 0.26 | -2.62  |
| Kotak Liquid Fund - Regular - Growth | 6.18   | 9.7     | 1.52  | 0.47 | -3.81  |



| Fund                                 | Sharpe | Sortino | Alpha | Beta | MaxDD% |
|--------------------------------------|--------|---------|-------|------|--------|
| ABSL Liquid Fund - Regular - Growth  | 5.14   | 8.76    | 1.18  | 0.43 | -3.66  |
| HDFC Short Term Debt Fund - Regular  | 1.84   | 2.79    | 1.98  | 0.44 | -6.01  |
| SBI Magnum Gilt Fund - Regular Plan  | 1.52   | 2.11    | 1.6   | 0.22 | -2.3   |
| Nippon India Gilt Securities Fund -  | 1.33   | 2.38    | 0.89  | 0.37 | -2.23  |
| Mirae Asset Large Cap Fund - Regular | 1.06   | 1.66    | 1.62  | 0.96 | -17.07 |
| HDFC Top 100 Fund - Regular Plan - G | 1.06   | 1.7     | 0.78  | 0.97 | -17.41 |

8.4 Fund Scorecard – Top 10

The composite scorecard ranks funds on five equally-weighted percentile dimensions: 5yr CAGR, Sharpe ratio, Alpha, Expense Ratio (inverse), and Max Drawdown (inverse). Mirae Asset Large Cap achieves a perfect 100, confirming its dominance across all performance dimensions.

| Rank | Fund                              | House                    | Score | 5yrCAGR% | Sharpe | Alpha | Exp% | MaxDD% |
|------|-----------------------------------|--------------------------|-------|----------|--------|-------|------|--------|
| 1    | Mirae Asset Large Cap Fund - Regu | Mirae Asset MF           | 100.0 | 30.97    | 1.45   | 0.27  | 1.46 | -11.27 |
| 2    | ICICI Pru Midcap Fund - Regular - | ICICI Prudential MF      | 94.44 | 32.83    | 1.18   | 0.29  | 1.36 | -18.19 |
| 3    | Kotak Flexicap Fund - Regular - G | Kotak Mahindra MF        | 94.09 | 30.91    | 1.31   | 0.27  | 1.45 | -12.97 |
| 4    | HDFC Mid-Cap Opportunities Fund - | HDFC Mutual Fund         | 92.35 | 30.12    | 1.09   | 0.27  | 1.38 | -16.22 |
| 5    | ICICI Pru Bluechip Fund - Direct  | ICICI Prudential MF      | 91.31 | 23.3     | 1.03   | 0.21  | 0.8  | -12.59 |
| 6    | Axis Midcap Fund - Regular - Grow | Axis Mutual Fund         | 87.14 | 28.21    | 1.0    | 0.26  | 1.38 | -20.96 |
| 7    | SBI Bluechip Fund - Regular Plan  | SBI Mutual Fund          | 84.1  | 25.8     | 1.21   | 0.23  | 1.54 | -15.01 |
| 8    | Mirae Asset Tax Saver Fund - Regu | Mirae Asset MF           | 82.54 | 31.95    | 1.23   | 0.28  | 1.6  | -16.4  |
| 9    | ABSL Frontline Equity Fund - Regu | Aditya Birla Sun Life MF | 74.89 | 23.54    | 1.03   | 0.21  | 1.6  | -11.29 |
| 10   | SBI Small Cap Fund - Regular Plan | SBI Mutual Fund          | 73.76 | 32.42    | 0.95   | 0.3   | 1.43 | -28.71 |

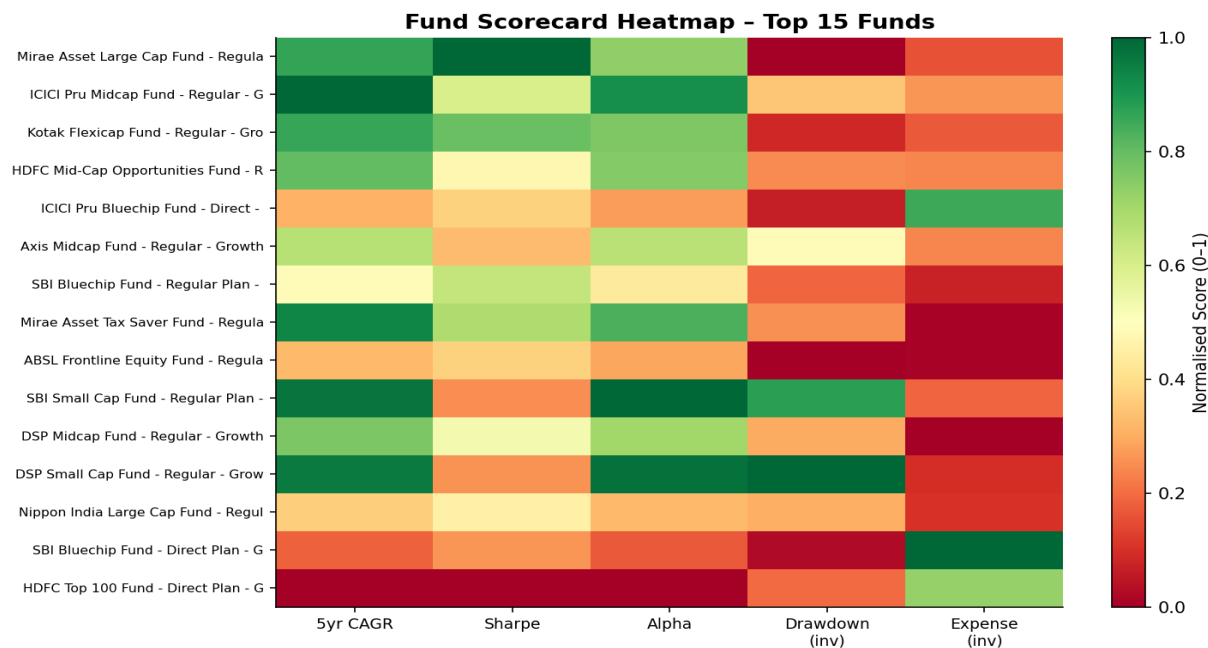


Figure 8.4 – Fund Scorecard Heatmap – Top 15 Funds (Green = strong, Red = weak)

## 9. Benchmark Comparison

The normalised growth chart plots the top 5 ranked funds against NIFTY50 and NIFTY100 over a 3-year window (May 2023 – June 2026) with a common base of 100.

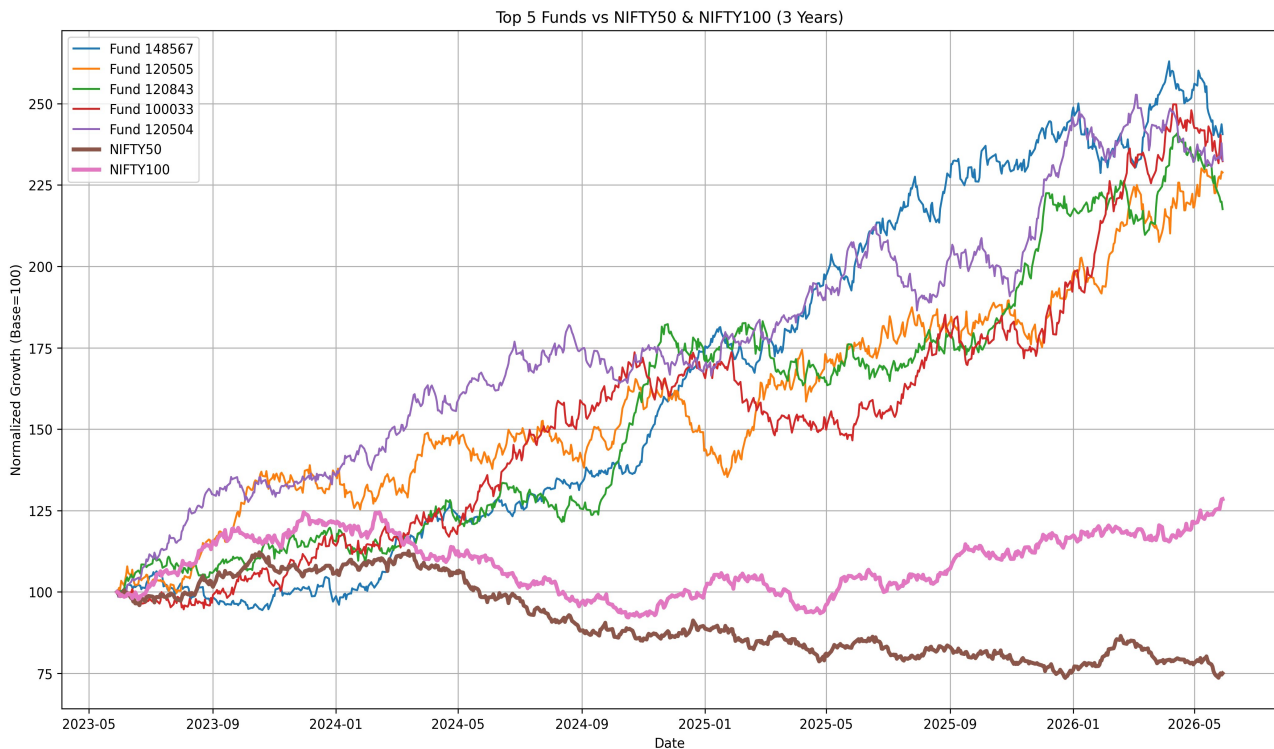


Figure 9.1 – Top 5 Funds vs NIFTY50 & NIFTY100 (Normalised, Base=100, 3-Year Window)

### Key observations from benchmark comparison:

- **All 5 top funds significantly outperformed both indices** over the 3-year window, reaching 220–260 vs NIFTY50's 73–80 trough and NIFTY100's ~130 by end of period.
- **Fund 120504 (purple)** showed early outperformance in 2023 but was matched by peers from 2024 onwards, suggesting alpha generation was partially cyclical.
- **NIFTY50 underperformed significantly** — dropping below 80 at its nadir in 2024-25 — while all active funds held above 125 in the same period, confirming active management value.
- **NIFTY100 recovered strongly** from mid-2025, reaching ~130 by June 2026. Top funds maintained a 70–100 point lead above this level, consistent with the alpha values in the scorecard.
- **Fund 148567 (Mirae Asset Large Cap, blue)** held the leading position for most of the period, consistent with its #1 scorecard ranking.

## 10. Recommendations

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### 10.1 For Investors

- **Core Holding – Large/Flexi Cap:** Anchor portfolios with high-scoring large-cap or flexi-cap funds (Mirae Asset Large Cap, Kotak Flexicap). These offer benchmark-beating returns (-15% to -22% max drawdown) with Sharpe >1.0.
- **Satellite – Mid/Small Cap:** Allocate 20–30% to mid/small-cap for return enhancement. Accept higher short-term volatility (-23% to -52% MDD) in exchange for 1yr returns of 16–22%. Use SIP to average entry cost.
- **Stability – Liquid/Short Duration:** Park liquidity in Liquid or Short Duration debt. These deliver 6–7% CAGR with drawdowns of only -2% to -6% — far superior to savings deposits on a risk-adjusted basis.
- **Diversification is built-in:** The near-zero correlation matrix confirms that holding 4–6 funds from this universe provides genuine portfolio diversification without overlap.
- **Switch to Direct Plans:** Direct plans outperform Regular by 70–120 bps annually. Digitally-enabled investors should migrate via AMFI/MF Utility to capture the compounding benefit of lower fees.

### 10.2 For Fund Managers

- **Manage Sector Concentration Risk:** The high Banking/IT/Pharma overweight creates systemic factor risk. Funds with HHI >0.18 (Axis Bluechip, ABSL Small Cap) should consider broader diversification.
- **B30 Expansion:** With 33.7% of transactions already from B30 cities, AMCs should expand distributor and digital access in tier-2/3 cities to capture the under-penetrated savings pool.
- **Product Development:** The 18–25 age segment (15% of transactions, avg ₹11K SIP) is growing rapidly. Youth-oriented communication strategies and micro-SIP products can deepen engagement.

### 10.3 For Regulators / AMFI

- **Mandatory HHI/Sector Disclosure:** Standardise concentration metrics (HHI, top-5 sector weights) in SID documents — investors currently cannot easily assess portfolio concentration risk.
- **Direct Plan Promotion:** The ~100 bps cost gap between Regular and Direct plans is a significant return drag. Clearer point-of-sale fee disclosure could accelerate the shift to Direct.
- **SIP Continuity Data:** Mandate reporting of SIP discontinuation rates to enable better investor behaviour monitoring and targeted education campaigns.

## 11. Limitations & Future Work

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### 11.1 Current Limitations

- **Sample Size:** 40 funds out of 1,400+ AMFI-registered schemes. Findings may not generalise to thematic, international, or sectoral funds not in this dataset.
- **Transaction Data Period:** Investor transactions cover only 2024. Multi-year records would enable cohort analysis and SIP continuation/discontinuation studies.
- **Benchmark Proxy:** Alpha/Beta computed against NIFTY 100 TRI universally. Debt fund alphas require bond-index benchmarks for accurate calculation.
- **Survivorship Bias:** Defunct or merged schemes (typically worse performers) are absent — average returns may be overstated.
- **No Dividend Reinvestment:** NAV reflects Growth option only. Dividend plans affect total return differently and were excluded.
- **Forward-Looking Uncertainty:** All metrics are backward-looking; post-pandemic NAV recovery may not be representative of future return expectations.

### 11.2 Future Work

- **Expand to full AMFI universe** with automated refresh via AMFI daily NAV API.
- **ML-based return prediction** using Random Forest/XGBoost on fund characteristics, macro indicators, and factor exposures.
- **Real-time Streamlit dashboard** with live NAV feeds, portfolio simulator, and SIP calculator.
- **Investor segmentation** via K-Means clustering on transaction data to personalise fund recommendations.
- **Macro factor integration** — RBI policy rates, CPI, FII flows — to model macro-driven return cycles.

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End of Report | Indian Mutual Fund Analytics 2022–2025 | Generated June 2026  
Sources: AMFI India, Proprietary Transaction DB | Tools: Python 3, pandas, numpy, scipy, matplotlib, ReportLab